

Supplementary Online Content

Chen Z, Brodie MJ, Liew D, Kwan P. Treatment outcomes in patients with newly diagnosed epilepsy treated with established and new antiepileptic drugs: a 30-year longitudinal cohort study. *JAMA Neurol*. Published online December 26, 2017. doi:10.1001/jamaneurol.2017.3949.

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This supplementary material has been provided by the authors to give readers additional information about their work.

eAppendix

“RULE-OF-THREE” SEIZURE-FREE RATE

According to the ILAE definition of drug resistant epilepsy,¹ 1,107 (61.7%) patients were seizure-free and 961 (53.5%) did so on monotherapy. Compared to one year seizure freedom, slightly fewer patients achieved ILAE defined seizure freedom on the first (795/1,795, 44.3%) and second (201/742, 27.1%) schedules (eTable 2). The seizure free rate was higher in patients with generalized epilepsy (251/386, 65.0%) compared with focal epilepsy (856/1,409, 60.8%) after adjustment for age at onset and gender ($p=0.036$).

Patients who did not achieve seizure freedom by the ILAE definition on the first AED had 1.70 times greater likelihood of developing uncontrolled epilepsy for each additional AED tried (odds ratio [OR], 1.70; 95% CI, 1.54 to 1.88), after adjusting for epilepsy classification, age at onset and gender. This was similar to the findings by using one year seizure freedom (OR 1.73; 95% CI, 1.56 to 1.91). There was also significant difference in overall probability of seizure freedom between patients treated with the first and second AED schedules ($p<0.001$; eFigure 3A). Similar effects were also observed in focal epilepsy (eFigure 3C), but the probability of seizure freedom was not significantly different between second and third AED schedules in generalized epilepsy ($p=0.13$; eFigure 3B).

Similar to one year seizure freedom, the effect of additional seizure control also diminished rapidly for each successive AED schedule when using the ILAE definition (eFigure 4). The second AED schedule could only provide an additional 10.9% probability of seizure freedom. After two AEDs were tried, the cumulative probabilities of seizure freedom were not significantly different with each successive AED schedule. Only 3.8% additional probability of seizure freedom was provided by the third AED schedule, when the first two schedules failed to control seizures. A total of 3.9% additional probability of seizure freedom was provided by the fourth or more AED schedules.

eReference

ONLINE-ONLY REFERENCES

1. Kwan P, Arzimanoglou A, Berg AT, et al. Definition of drug resistant epilepsy: consensus proposal by the ad hoc Task Force of the ILAE Commission on Therapeutic Strategies. *Epilepsia*. 2010;51(6):1069-1077.

ONLINE-ONLY TABLES

eTable 1. Demographics and treatment outcome of patients recruited in different time periods.

Recruitment Period		1982-1991 n=142		1992-2001 n=684		2002-2012 n=969	
Demographics							
Sex (%)							
	Female	71	(50)	322	(47)	433	(45)
	Male	71	(50)	362	(53)	536	(55)
Age at onset (years)							
	Mean (SD)	30	(18)	36	(20)	36	(18)
	Median (IQR)	23	(17-39)	30	(20-49)	33	(20-49)
	Range	5-80		1-92		1-87	
Age at referral (years)							
	Mean (SD)	32	(18)	36	(20)	38	(18)
	Median (IQR)	26	(19-42)	31	(20-49)	35	(22-51)
	Range	12-80		9-93		12-87	
Type of epilepsy (%)							
	Generalised	51	(36)	142	(21)	193	(20)
	Focal	91	(64)	542	(79)	776	(80)
Treatment Outcome (%)*							
	Seizure-free**	89	(63)	437	(64)	591	(61)
	Refractory	53	(37)	247	(36)	378	(39)

*Treatment outcome was assessed at the last follow up after minimum of two years from treatment initiation, i.e. at the end of 1993 for patients started the first AED between 1982 and 1991, at 2003 for those who started treatment between 1992 and 2001, and at 2014 for those started AED between 2002 and 2012. ** $p=0.49$ before and $p=0.43$ after adjusting for demographics (Chi-square test).

eTable 2. "Rule-of-Three" seizure-free rates with successive antiepileptic drug schedules

AED Schedule	Number of patients tried the AED schedule	Patients Achieved Seizure-freedom			
		n	% of patients tried the AED schedule	% of 1,107 patients achieved Seizure-freedom	% of total 1,795 patients
1	1,795	795	44.3%	71.8%	44.3%
2	742	201	27.1%	18.2%	11.2%
3	330	77	23.3%	6.96%	4.29%
4	140	19	13.6%	1.72%	1.06%
5	71	9	12.7%	0.81%	0.50%
6	43	5	11.6%	0.45%	0.28%
7	15	1	6.67%	0.09%	0.06%
8	9	0	0%	0%	0%
9	5	0	0%	0%	0%
10	2	0	0%	0%	0%
11	1	0	0%	0%	0%

AED, antiepileptic drug

ONLINE-ONLY FIGURE LEGENDS

eFigure 1. Antiepileptic drug prescriptions over the study period for antiepileptic drugs prescribed in subsequent treatment schedules.

The colored areas represent the prescription of each antiepileptic drug as a proportion of all the antiepileptic drugs prescribed in the corresponding years. The yellow dash lines divide the established and new AEDs. Established AEDs: CBZ, carbamazepine; VPA, valproic acid; PHT, phenytoin; CLB, clobazam. New AEDs: LTG, lamotrigine; OXC, oxcarbazepine; GBP, gabapentin; LEV, levetiracetam; TPM, topiramate; ZNS, zonisamide; PGB, pregabalin; LCM, lacosamide. Other AEDs include vigabatrin, felbamate, tiagabine, rufinamide, eslicarbazepine, retigabine, perampanel and unnamed trial drugs.

eFigure 2. Cumulative probability of one-year seizure freedom by time from start of treatment in the three subgroups. The solid navy blue, long dash maroon and dash-dot green lines represent the probabilities of achieving one-year seizure in patients who commenced treatment in 1982-1991, 1992-2001 and 2002-2012, respectively. (log-rank test, $p=0.47$).

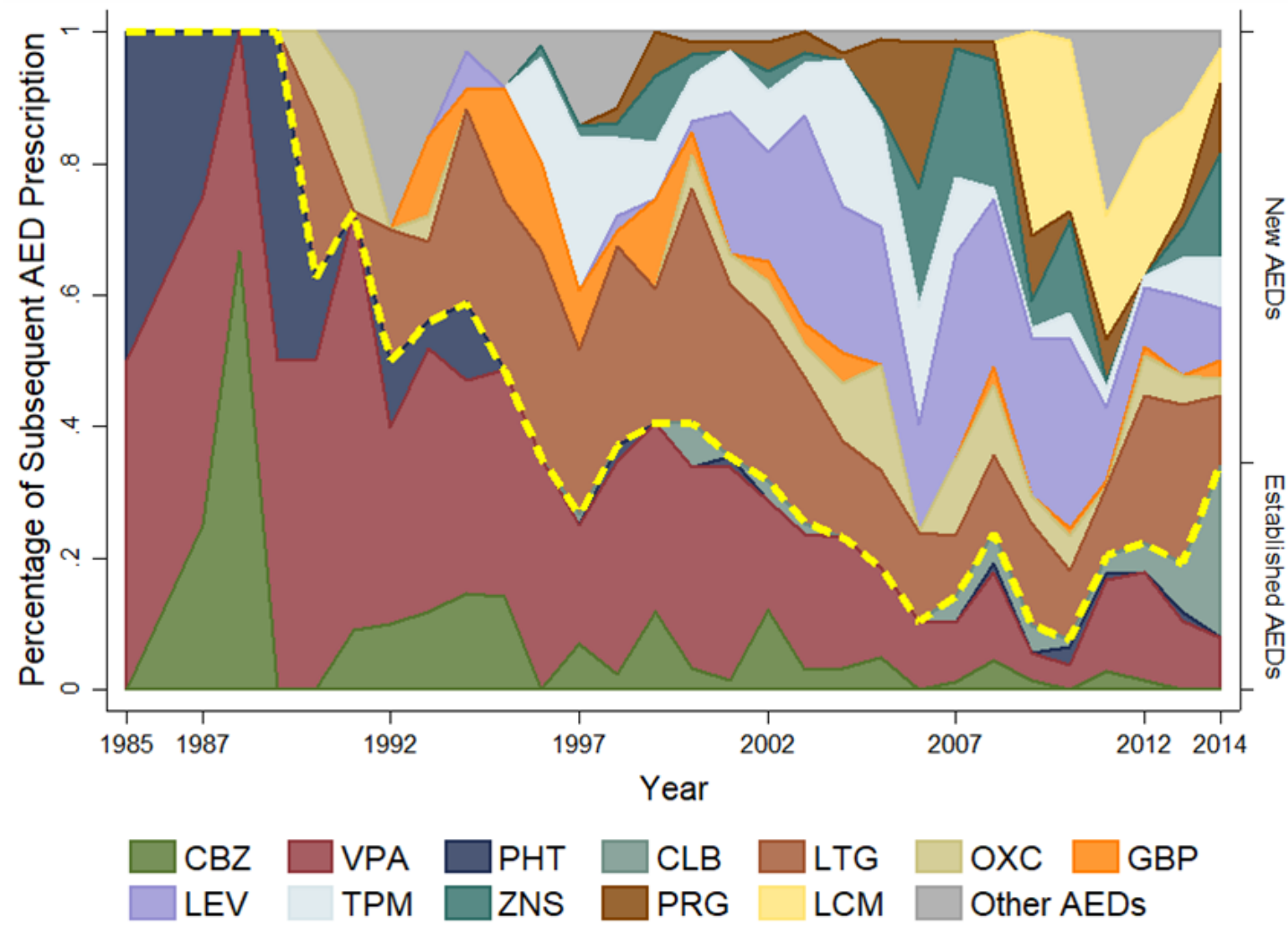
eFigure 3. Cumulative probability of “Rule-of-Three” seizure freedom by time from start of treatment and number of antiepileptic drug tried for (A) all patients; (B) patients with generalised epilepsies; and (C) patients with focal epilepsies.

The solid navy blue, long dash maroon, dash-dot green, short dash orange and dotted blue lines represent the probabilities of achieving “Rule-of-Three” seizure freedom after certain months of treatment on the first, second, third, fourth, and fifth or above antiepileptic drug schedules, respectively. Owing to the small sample size, patients who tried more than three antiepileptic drugs were grouped together in the separate analysis for (B) patients with generalised epilepsy and (C) patients with focal epilepsy.

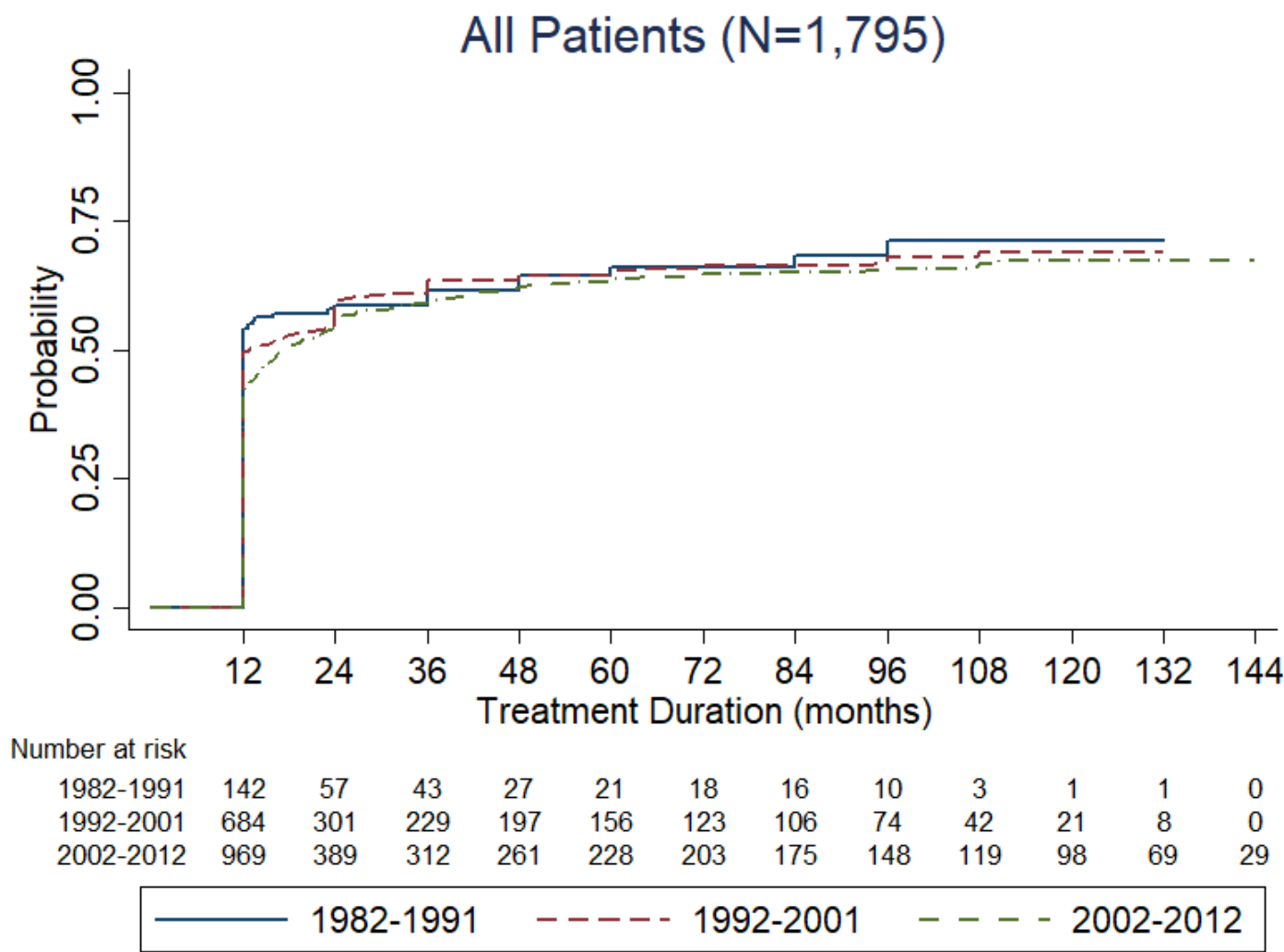
eFigure 4. Additional effects on probability of “Rule-of-Three” seizure freedom for each antiepileptic drug schedule.

The white area of each stacked bar represents the probability to achieve “Rule-of-Three” seizure freedom at the current antiepileptic drug schedule and the rate is displayed next to each bar. The grey area represents the cumulative seizure-free rate obtained on the previous antiepileptic drug schedule(s).

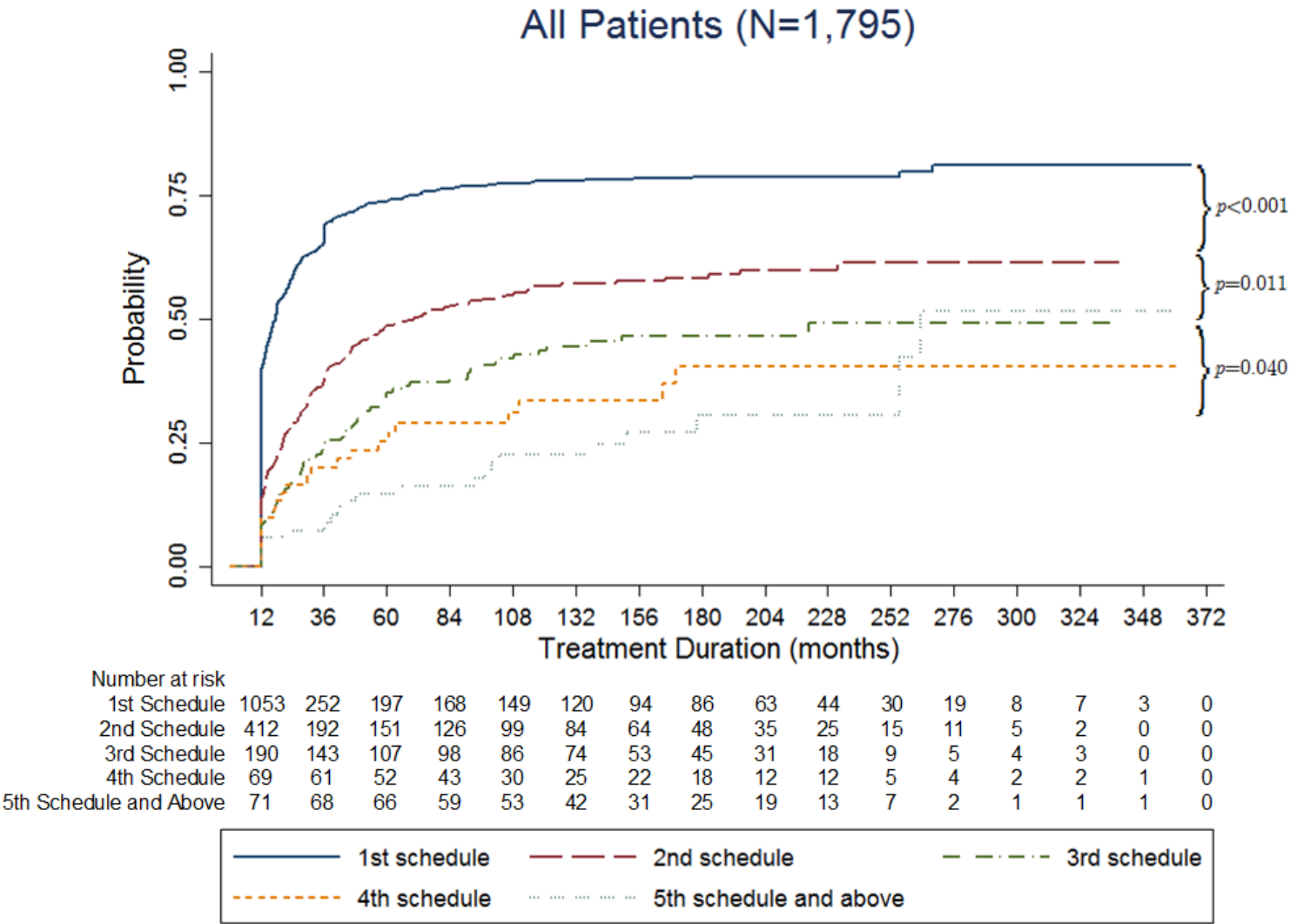
eFIGURE 1



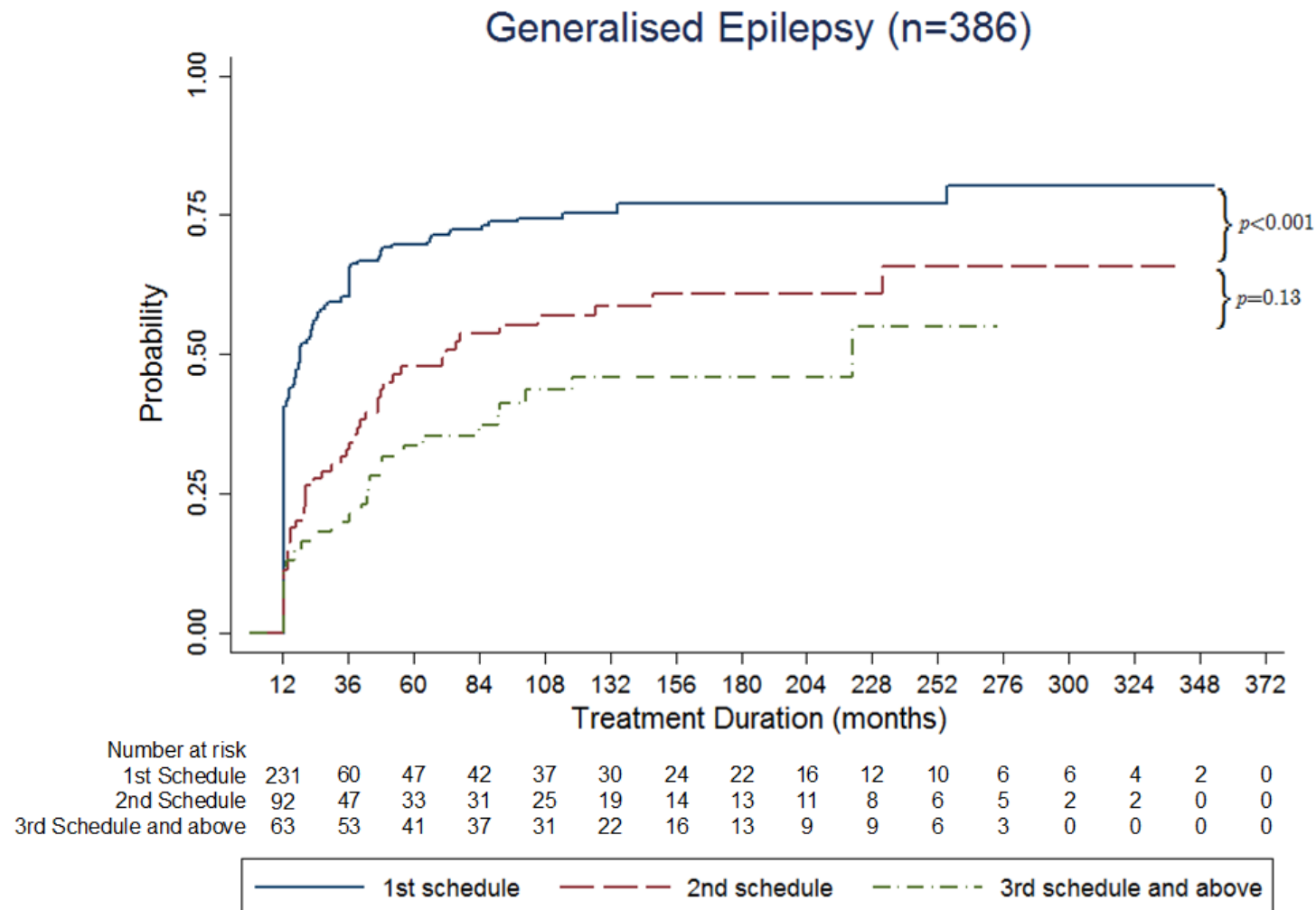
eFIGURE 2



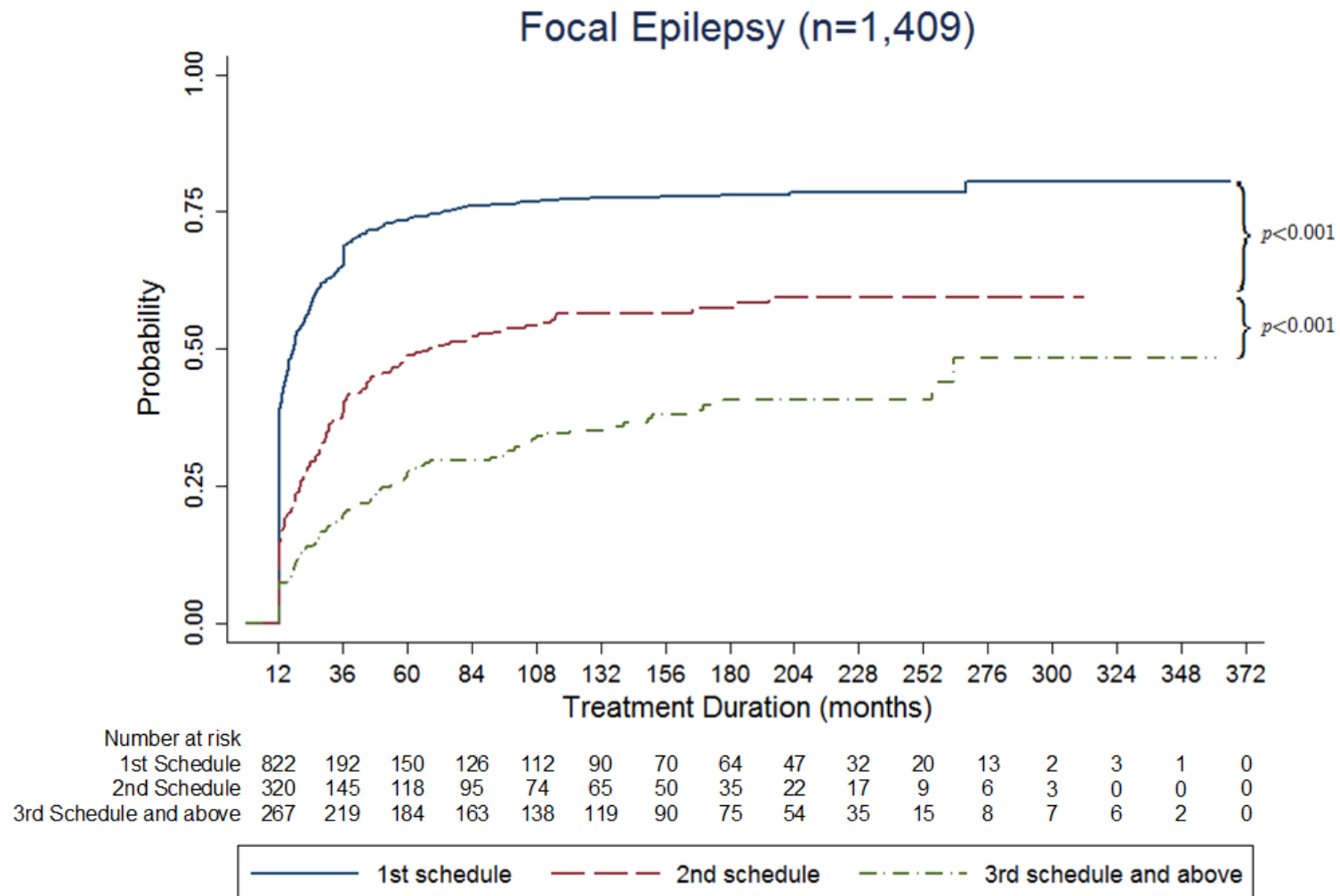
eFIGURE 3
A. All patients (N=1,795)



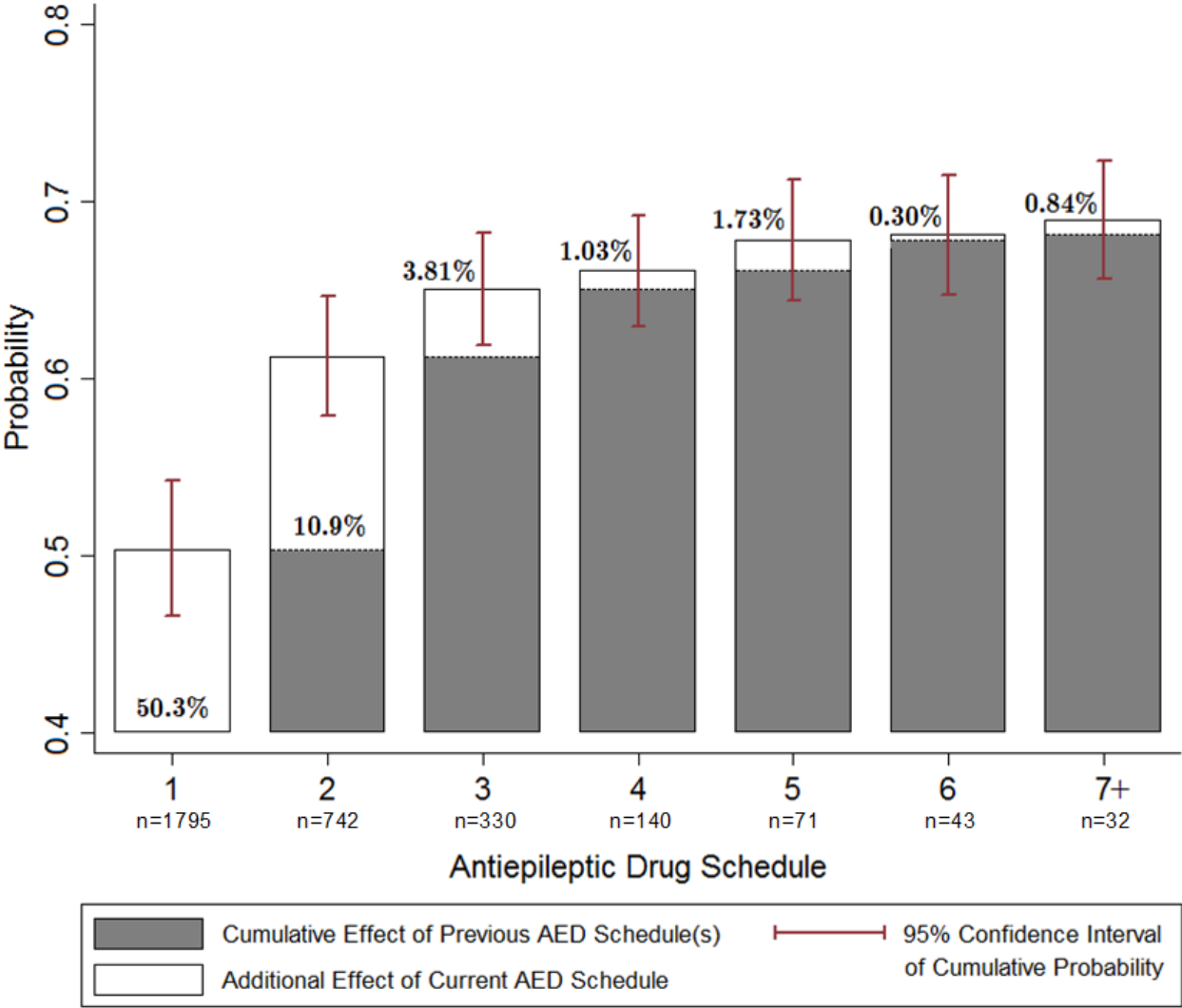
B. Patients with generalized epilepsies (n=386)



C. Patients with focal epilepsies (n=1,409)



eFIGURE 4



eTable 3 Associations between terminal seizure outcome and individual potential risk factors after adjustment for gender and epilepsy classification, univariate analysis

	Odds ratio	95% CI	<i>p</i>-value
One year pretreatment seizure number	0.97	(0.96-0.99)	<0.001
Smoking	0.44	(0.29-0.69)	<0.001
Recreational drug use	0.42	0.24-0.73	0.002
Family history of epilepsy – first degree	0.60	0.47-0.79	<0.001
History of brain injury	0.72	0.54-0.96	0.027
History of brain infection	0.93	0.52-1.68	0.81
History of febrile seizures	1.05	0.67-1.65	0.82
History of birth trauma	1.01	0.76-1.3	0.96
Psychiatric comorbidity	0.45	0.30-0.6	<0.001

eTable 4 Associations between terminal seizure outcome and potential risk factors after adjustment for sex and epilepsy classification

	Odds ratio	95% CI	<i>p</i>-value
One year pretreatment seizure number	0.94	(0.89-0.98)	0.007
Recreational drug use	0.36	(0.20-0.66)	0.001
Family history of epilepsy – first degree	0.45	(0.27-0.78)	0.004